

在 gRPC Go 中設定基本 OpenTelemetry 外掛程式

來源：<https://codelabs.developers.google.com/grpc/basic-otel-plugin-grpc-go?hl=zh-tw>

步驟數：6

如何為 gRPC Go 應用程式設定 OpenTelemetry 外掛程式

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1. 簡介

在本程式碼研究室中，您將使用 gRPC 建立用戶端和伺服器，做為以 Go 編寫的路線對應應用程式基礎。

在本教學課程結束時，您將擁有一個簡單的 gRPC HelloWorld 應用程式，並使用 gRPC OpenTelemetry 外掛程式進行檢測，且能夠在 Prometheus 中查看匯出的可觀測性指標。

課程內容

- 如何為現有的 gRPC Go 應用程式設定 OpenTelemetry 外掛程式
 - 執行本機 Prometheus 執行個體
 - 將指標匯出至 Prometheus
 - 從 Prometheus 資訊主頁查看指標
-

2. 事前準備

軟硬體需求

- `git`
- `curl`
- Go [Go 最新兩個主要版本](#)的其中一個。
- `protoc` v3

設定前置條件：

```
sudo apt-get update -y
sudo apt-get upgrade -y
sudo apt-get install -y git curl
go install google.golang.org/protobuf/cmd/protoc-gen-go@latest
go install google.golang.org/grpc/cmd/protoc-gen-go-grpc@latest
```

取得程式碼

為簡化學習過程，本程式碼研究室提供預先建構的原始碼架構，協助您踏出第一步。下列步驟會引導您在應用程式中檢測 gRPC OpenTelemetry 外掛程式。

grpc-codelabs

本程式碼研究室的架構原始碼位於這個 GitHub [目錄](#)。如果您不想自行導入程式碼，可以前往「completed」目錄查看已完成的原始碼。

首先，請複製 grpc Codelab 存放區，然後 cd 到 grpc-go-opentelemetry 資料夾：

□□□

```
git clone https://github.com/grpc-ecosystem/grpc-codelabs.git
cd grpc-codelabs/codelabs/grpc-go-opentelemetry/
```

或者，您也可以下載只包含 Codelab 目錄的 .zip 檔案，然後手動解壓縮。

步驟 3 · 約 7 分鐘

3. 註冊 OpenTelemetry 外掛程式

我們需要 gRPC 應用程式來新增 gRPC OpenTelemetry 外掛程式。在本程式碼研究室中，我們將使用簡單的 gRPC HelloWorld 用戶端和伺服器，並透過 gRPC OpenTelemetry 外掛程式進行檢測。

在用戶端設定插樁

首先，請在用戶端中註冊已設定 Prometheus 匯出器的 OpenTelemetry 外掛程式。使用您慣用的編輯器開啟 `codelabs/grpc-go-opentelemetry/start_here/client/main.go`，然後轉換 `main()`，加入程式碼來設定 gRPC Go Otel 外掛程式。

建立新的 Prometheus 匯出工具

下列程式碼片段會建立新的 [Prometheus 匯出工具](#)

```
exporter, err := prometheus.New()
if err != nil {
    log.Fatalf("Failed to start prometheus exporter: %v", err)
}
```

注意：系統會在 OpenTelemetry Meter Provider 上設定 Prometheus Exporter。(您也可以透過其他方式匯出指標。本程式碼研究室選擇使用 prometheus 匯出工具。) 這個 MeterProvider 會提供給 gRPC 的 OpenTelemetry 外掛程式做為設定。OpenTelemetry 外掛程式完成全域註冊後，所有 gRPC 用戶端和伺服器都會檢測 OpenTelemetry。

啟動 Prometheus 匯出工具

啟動新的 goroutine，使用 `net/http` 在 `prometheus_endpoint` 旗標傳入的位址接聽及提供服務

```
go http.ListenAndServe(*prometheusEndpoint, promhttp.Handler())
```

使用 OpenTelemetry 外掛程式建立 DialOption

使用 gRPC Otel [DialOption](#) 方法，透過 Prometheus Meter Provider 設定 DialOption

```
provider := metric.NewMeterProvider(metric.WithReader(exporter))
do := opentelemetry.DialOption(opentelemetry.Options{MetricsOptions:
    opentelemetry.MetricsOptions{MeterProvider: provider}})
```

連線至伺服器時傳遞 DialOption

將 `do` 傳遞至 `gRPC.NewClient`

這個 MeterProvider 會提供給 gRPC 的 OpenTelemetry 外掛程式做為設定。OpenTelemetry 外掛程式完成全域註冊後，所有 gRPC 用戶端和伺服器都會檢測 OpenTelemetry。

在伺服器上設定檢測

同樣地，我們也將 OpenTelemetry 外掛程式新增至伺服器。開啟 `codelabs/grpc-go-opentelemetry/start_here/server/main.go` 並編寫程式碼，設定 gRPC Go Otel 外掛程式

建立新的 Prometheus 匯出工具

```
exporter, err := prometheus.New()
if err != nil {
    log.Fatalf("Failed to start prometheus exporter: %v", err)
}
```

使用 Prometheus 匯出工具建立新的電錶供應商

```
provider := metric.NewMeterProvider(metric.WithReader(exporter))
go http.ListenAndServe(*prometheusEndpoint, promhttp.Handler())

so := opentelemetry.ServerOption(opentelemetry.Options{MetricsOptions:
    opentelemetry.MetricsOptions{MeterProvider: provider}})
```

將這個伺服器選項傳遞至 `gRPC.NewServer`

步驟 4 · 約 0 分鐘

4. 執行範例並查看指標

如要執行伺服器，請執行下列指令：

```
cd server
go mod tidy
go run .
```

設定成功後，伺服器會顯示下列輸出內容：

```
Serving on :50051
```

伺服器執行期間，在另一個終端機中執行用戶端：

```
cd client
go mod tidy
go run .
```

成功執行的結果如下所示：

```
message:"this is examples/opentelemetry (from :50051)"
message:"this is examples/opentelemetry (from :50051)"
message:"this is examples/opentelemetry (from :50051)"
```

我們已設定 gRPC OpenTelemetry 外掛程式，使用 Prometheus 匯出指標。這些指標會顯示在伺服器的 localhost:9464 和用戶端的 localhost:9465。

如要查看用戶端指標，請按照下列步驟操作：

```
curl localhost:9465/metrics
```

結果會採用以下形式：

```
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in
garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100.
This value is set by the GOGC environment variable, and the runtime/debug.SetGCPercent
function. Sourced from /gc/gogc:percent
# TYPE go_gc_gogc_percent gauge
go_gc_gogc_percent 100
# HELP go_gc_gomemlimit_bytes Go runtime memory limit configured by the user, otherwise
math.MaxInt64. This value is set by the GOMEMLIMIT environment variable, and the
runtime/debug.SetMemoryLimit function. Sourced from /gc/gomemlimit:bytes
# TYPE go_gc_gomemlimit_bytes gauge
go_gc_gomemlimit_bytes 9.223372036854776e+18
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 14
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.22.1"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated in heap and currently in use. Equals
to /memory/classes/heap/objects:bytes.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 662176
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated in heap until now, even
if released already. Equals to /gc/heap/allocs:bytes.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 662176
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash
table. Equals to /memory/classes/profiling/buckets:bytes.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 8739
# HELP go_memstats_frees_total Total number of heap objects frees. Equals to
/gc/heap/frees:objects + /gc/heap/tiny/allocs:objects.
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 0
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata.
Equals to /memory/classes/metadata/other:bytes.
# TYPE go_memstats_gc_sys_bytes gauge
go_memstats_gc_sys_bytes 1.554232e+06
# HELP go_memstats_heap_alloc_bytes Number of heap bytes allocated and currently in use, same
as go_memstats_alloc_bytes. Equals to /memory/classes/heap/objects:bytes.
# TYPE go_memstats_heap_alloc_bytes gauge
go_memstats_heap_alloc_bytes 662176
# HELP go_memstats_heap_idle_bytes Number of heap bytes waiting to be used. Equals to
/memory/classes/heap/released:bytes + /memory/classes/heap/free:bytes.
```



```
# TYPE go_memstats_heap_idle_bytes gauge
go_memstats_heap_idle_bytes 4.75136e+06
# HELP go_memstats_heap_inuse_bytes Number of heap bytes that are in use. Equals to
/memory/classes/heap/objects:bytes + /memory/classes/heap/unused:bytes
# TYPE go_memstats_heap_inuse_bytes gauge
go_memstats_heap_inuse_bytes 2.981888e+06
# HELP go_memstats_heap_objects Number of currently allocated objects. Equals to
/gc/heap/objects:objects.
# TYPE go_memstats_heap_objects gauge
go_memstats_heap_objects 4022
# HELP go_memstats_heap_released_bytes Number of heap bytes released to OS. Equals to
/memory/classes/heap/released:bytes.
# TYPE go_memstats_heap_released_bytes gauge
go_memstats_heap_released_bytes 4.75136e+06
# HELP go_memstats_heap_sys_bytes Number of heap bytes obtained from system. Equals to
/memory/classes/heap/objects:bytes + /memory/classes/heap/unused:bytes +
/memory/classes/heap/released:bytes + /memory/classes/heap/free:bytes.
# TYPE go_memstats_heap_sys_bytes gauge
go_memstats_heap_sys_bytes 7.733248e+06
# HELP go_memstats_last_gc_time_seconds Number of seconds since 1970 of last garbage
collection.
# TYPE go_memstats_last_gc_time_seconds gauge
go_memstats_last_gc_time_seconds 0
# HELP go_memstats_mallocs_total Total number of heap objects allocated, both live and gc-ed.
Semantically a counter version for go_memstats_heap_objects gauge. Equals to
/gc/heap/allocs:objects + /gc/heap/tiny/allocs:objects.
# TYPE go_memstats_mallocs_total counter
go_memstats_mallocs_total 4022
# HELP go_memstats_mcache_inuse_bytes Number of bytes in use by mcache structures. Equals to
/memory/classes/metadata/mcache/inuse:bytes.
# TYPE go_memstats_mcache_inuse_bytes gauge
go_memstats_mcache_inuse_bytes 9600
# HELP go_memstats_mcache_sys_bytes Number of bytes used for mcache structures obtained from
system. Equals to /memory/classes/metadata/mcache/inuse:bytes +
/memory/classes/metadata/mcache/free:bytes.
# TYPE go_memstats_mcache_sys_bytes gauge
go_memstats_mcache_sys_bytes 15600
# HELP go_memstats_mspan_inuse_bytes Number of bytes in use by mspan structures. Equals to
/memory/classes/metadata/mspan/inuse:bytes.
# TYPE go_memstats_mspan_inuse_bytes gauge
go_memstats_mspan_inuse_bytes 65440
# HELP go_memstats_mspan_sys_bytes Number of bytes used for mspan structures obtained from
system. Equals to /memory/classes/metadata/mspan/inuse:bytes +
/memory/classes/metadata/mspan/free:bytes.
# TYPE go_memstats_mspan_sys_bytes gauge
go_memstats_mspan_sys_bytes 81600
# HELP go_memstats_next_gc_bytes Number of heap bytes when next garbage collection will take
place. Equals to /gc/heap/goal:bytes.
# TYPE go_memstats_next_gc_bytes gauge
go_memstats_next_gc_bytes 4.194304e+06
# HELP go_memstats_other_sys_bytes Number of bytes used for other system allocations. Equals
to /memory/classes/other:bytes.
```

```

# TYPE go_memstats_other_sys_bytes gauge
go_memstats_other_sys_bytes 1.111813e+06
# HELP go_memstats_stack_inuse_bytes Number of bytes obtained from system for stack allocator
in non-CGO environments. Equals to /memory/classes/heap/stacks:bytes.
# TYPE go_memstats_stack_inuse_bytes gauge
go_memstats_stack_inuse_bytes 655360
# HELP go_memstats_stack_sys_bytes Number of bytes obtained from system for stack allocator.
Equals to /memory/classes/heap/stacks:bytes + /memory/classes/os-stacks:bytes.
# TYPE go_memstats_stack_sys_bytes gauge
go_memstats_stack_sys_bytes 655360
# HELP go_memstats_sys_bytes Number of bytes obtained from system. Equals to
/memory/classes/total:byte.
# TYPE go_memstats_sys_bytes gauge
go_memstats_sys_bytes 1.1160592e+07
# HELP go_sched_gomaxprocs_threads The current runtime.GOMAXPROCS setting, or the number of
operating system threads that can execute user-level Go code simultaneously. Sourced from
/sched/gomaxprocs:threads
# TYPE go_sched_gomaxprocs_threads gauge
go_sched_gomaxprocs_threads 8
# HELP go_threads Number of OS threads created.
# TYPE go_threads gauge
go_threads 8
# HELP grpc_client_attempt_duration_seconds End-to-end time taken to complete a client call
attempt.
# TYPE grpc_client_attempt_duration_seconds histogram
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1e-05"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="5e-05"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.0001"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.0003"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.0006"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.0008"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.001"} 0
grpc_client_attempt_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",g
rpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.002"} 0

```

[illegible]

[illegible]

```

go",otel_scope_version="1.67.0-dev"} 26
# HELP grpc_client_attempt_rcvd_total_compressed_message_size_bytes Compressed message bytes
received per call attempt.
# TYPE grpc_client_attempt_rcvd_total_compressed_message_size_bytes histogram
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0"} 0
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1024"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="2048"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="4096"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="16384"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="65536"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="262144"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1.048576e+06"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="4.194304e+06"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1.6777216e+07"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="6.7108864e+07"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="2.68435456e+08"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1.073741824e+09"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="4.294967296e+09"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="+Inf"} 26
grpc_client_attempt_rcvd_total_compressed_message_size_bytes_sum{grpc_method="grpc.example.e
cho.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev"} 1196

```



```

grpc_client_attempt_rcvd_total_compressed_message_size_bytes_count{grpc_method="grpc.examples
.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev"} 26
# HELP grpc_client_attempt_sent_total_compressed_message_size_bytes Compressed message bytes
sent per client call attempt.
# TYPE grpc_client_attempt_sent_total_compressed_message_size_bytes histogram
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0"} 0
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1024"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="2048"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="4096"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="16384"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="65536"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="262144"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1.048576e+06"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="4.194304e+06"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1.6777216e+07"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="6.7108864e+07"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="2.68435456e+08"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1.073741824e+09"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="4.294967296e+09"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_bucket{grpc_method="grpc.example
s.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="+Inf"} 26
grpc_client_attempt_sent_total_compressed_message_size_bytes_sum{grpc_method="grpc.examples.e

```

```
cho.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev"} 832
grpc_client_attempt_sent_total_compressed_message_size_bytes_count{grpc_method="grpc.examples
.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev"} 26
# HELP grpc_client_attempt_started_total Number of client call attempts started.
# TYPE grpc_client_attempt_started_total counter
grpc_client_attempt_started_total{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev"} 27
# HELP grpc_client_call_duration_seconds Time taken by gRPC to complete an RPC from application's perspective.
# TYPE grpc_client_call_duration_seconds histogram
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="1e-05"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="5e-05"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.0001"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.0003"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.0006"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.0008"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.001"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.002"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.003"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.004"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.005"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev",le="0.006"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
```

[illegible]


```

go",otel_scope_version="1.67.0-dev",le="0.4"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.5"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.65"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="0.8"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="1"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="2"} 0
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="5"} 26
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="10"} 26
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="20"} 26
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="50"} 26
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="100"} 26
grpc_client_call_duration_seconds_bucket{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc
_status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-
go",otel_scope_version="1.67.0-dev",le="+Inf"} 26
grpc_client_call_duration_seconds_sum{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_st
atus="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-
dev"} 52.35282954500001
grpc_client_call_duration_seconds_count{grpc_method="grpc.examples.echo.Echo/UnaryEcho",grpc_
status="OK",grpc_target="dns:///50051",otel_scope_name="grpc-go",otel_scope_version="1.67.0-
dev"} 26
# HELP otel_scope_info Instrumentation Scope metadata
# TYPE otel_scope_info gauge
otel_scope_info{otel_scope_name="grpc-go",otel_scope_version="1.67.0-dev"} 1
# HELP promhttp_metric_handler_requests_in_flight Current number of scrapes being served.
# TYPE promhttp_metric_handler_requests_in_flight gauge
promhttp_metric_handler_requests_in_flight 1
# HELP promhttp_metric_handler_requests_total Total number of scrapes by HTTP status code.
# TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 1
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0
# HELP target_info Target metadata

```

```
# TYPE target_info gauge
target_info{service_name="unknown_service:client",telemetry_sdk_language="go",telemetry_sdk_name="opentelemetry",telemetry_sdk_version="1.28.0"} 1
```

同樣地，伺服器端指標也是如此：

```
curl localhost:9464/metrics
```

步驟 5 · 約 5 分鐘

5. 在 Prometheus 中查看指標

在這裡，我們將設定 Prometheus 執行個體，用於擷取 gRPC 範例用戶端和伺服器，這些用戶端和伺服器會使用 Prometheus 匯出指標。

下載適用於您平台的最新版 [Prometheus](#)，然後解壓縮並執行：

```
tar xvfz prometheus-*.tar.gz
cd prometheus-*
```

建立 Prometheus 設定檔，並加入下列內容：

```
cat > grpc_otel_go_prometheus.yml <<EOF
scrape_configs:
  - job_name: "prometheus"
    scrape_interval: 5s
    static_configs:
      - targets: ["localhost:9090"]
  - job_name: "grpc-otel-go"
    scrape_interval: 5s
    static_configs:
      - targets: ["localhost:9464", "localhost:9465"]
EOF
```

使用新設定啟動 Prometheus：

```
./prometheus --config.file=grpc_otel_go_prometheus.yml
```

這會設定每 5 秒擷取一次用戶端和伺服器程式碼實驗室程序的指標。

前往 <http://localhost:9090/graph> 查看指標。例如，查詢 -

```
histogram_quantile(0.5, rate(grpc_client_attempt_duration_seconds_bucket[1m]))
```

會顯示圖表，其中以 1 分鐘做為分位數計算的時間視窗，顯示嘗試延遲時間中位數。

查詢率 -

```
increase(grpc_client_attempt_duration_seconds_bucket[1m])
```

步驟 6 · 約 5 分鐘

6. (選用) 使用者練習

在 Prometheus 資訊主頁中，您會發現 QPS 偏低。看看您是否能在範例中找出限制 QPS 的可疑程式碼。

對於熱衷於此的開發人員，用戶端程式碼會限制在特定時間點只能有一個待處理的 RPC。您可以修改這項設定，讓用戶端傳送更多 RPC，不必等待先前的 RPC 完成。(這項問題的解決方案尚未提供)。
